Handwritten Digit Generation Using Deep Convolutional Generative Adversarial Networks on MNIST Dataset

1st Harsh Shukla
Computer Engineering
Marwadi University
Rajkot, Gujarat, India
Harsh.shukla132525@marwadiuniversity.ac.in

2st Darshak Ramani
Computer Engineering
Marwadi University
Rajkot, Gujarat, India
darshak.ramani130904@marwadiuniversity.ac.in

3st Shyam Patel
Computer Engineering
Marwadi University
Rajkot, Gujarat, India
Shyam.patel131866@marwadiuniversity.ac.in

4st Nitin Varshney
Computer Engineering
Marwadi University
Rajkot, Gujarat, India
nitin.varshney@marwadiuniversity.edu.in

Abstract

This paper presents a comprehensive implementation and evaluation of a Deep Convolutional Generative Adversarial Network for synthesizing realistic handwritten digit images. Using the MNIST benchmark dataset comprising 60,000 training samples, our DCGAN model trains adversarially for 100 epochs with carefully tuned hyperparameters including batch size 128, learning rate 2×10^{-4}, and label smoothing at 0.9. The trained generator achieves a Fréchet Inception Distance of 5.94 against 10,000 real samples and produces all ten digit classes with distribution coverage between 7.7% and 13.4%. A downstream CNN classifier trained exclusively on generated images achieves 87.83% accuracy on real MNIST test data, validating practical utility for data augmentation. Our findings demonstrate that a well tuned DCGAN without complex loss modifications can achieve FID scores competitive with WGAN GP variants while maintaining architectural simplicity and fast inference suitable for resource constrained deployment scenarios.

Index Terms Generative Adversarial Networks, DCGAN, MNIST, Handwritten Digit Generation, Deep

Learning, FID Score, Data Augmentation

I. Introduction

Handwritten digit recognition remains a foundational problem in computer vision with practical applications spanning banking cheque processing, postal automation, and document digitization. While discriminative models for digit classification have achieved near perfect accuracy, the complementary task of generating realistic synthetic digits presents unique challenges and opportunities. Synthetic digit generation enables data augmentation for low resource scenarios, privacy preserving model training, and creative applications in content generation.

Generative Adversarial Networks, introduced by Goodfellow et al. in 2014, provide an elegant framework for learning complex data distributions through adversarial competition. The architecture pits a Generator network that creates synthetic samples against a Discriminator network that attempts to distinguish real from generated data. Through iterative training, both networks improve until the Generator produces samples indistinguishable from authentic data.

Despite the theoretical appeal of GANs, practical implementation faces substantial hurdles. Training instability, where losses oscillate without convergence, and mode collapse, where the Generator produces only a subset of possible outputs, are well documented challenges. The DCGAN architecture proposed by Radford et al. in 2015 addressed many stability issues through convolutional layers, batch normalization, and LeakyReLU activations. However, even DCGAN implementations frequently exhibit partial mode collapse on MNIST, with certain digit classes underrepresented in generated outputs.

Recent literature from 2021 onwards has explored various enhancements to GAN based digit generation. A 2022 study demonstrated parameter adjustment through multiple training rounds to improve generation robustness. Comparative analysis published in 2024 evaluated Simple GAN, DCGAN, SNGAN, and WGAN GP on MNIST, finding that SNGAN offered superior training stability while WGAN GP provided competitive image quality. A 2025 study proposed Coordinate Attention DCGAN for improved performance under limited data conditions.

The contribution of this paper is threefold. First, we provide a rigorously documented DCGAN implementation with comprehensive quantitative evaluation using industry standard metrics including FID computed over 10,000 samples. Second, we validate the practical utility of generated digits through downstream classifier training, achieving 87.83% accuracy on real test data. Third, we demonstrate that a carefully tuned standard DCGAN can achieve FID scores competitive with more complex architectures while maintaining implementation accessibility.

II. Related Work

A. Evolution of GAN Architectures

The GAN framework established adversarial training as a viable approach to generative modeling but offered limited architectural guidance. Radford et al. addressed this gap with DCGAN, introducing convolutional layers, batch normalization, and specific activation functions that dramatically improved training stability and output quality. These architectural principles remain foundational to most image generation work.

Mirza and Osindero introduced Conditional GANs, enabling targeted generation through class label conditioning. Odena et al. refined this approach with Auxiliary Classifier GANs, where the Discriminator jointly predicts authenticity and class membership. While these conditional variants offer greater control, they require fully labeled training data and introduce additional training complexity.

B. Handwritten Digit Generation Studies (2021 2025)

Several recent studies have specifically addressed handwritten digit generation using GANs on MNIST. A 2022 IEEE publication demonstrated that GAN models achieve superior accuracy through network parameter adjustments across multiple training rounds. This work validates the fundamental approach but uses basic GAN architecture without convolutional improvements.

A 2023 study combined ESRGAN with CNN classifiers on MNIST, using adversarial training to produce enhanced resolution digit images. The work showed that super resolution GANs can improve output quality beyond standard generation but at increased computational cost.

A comprehensive comparative study published in 2024 evaluated four GAN variants on MNIST. The analysis revealed that Spectral Normalization GAN offered the most stable training process, while WGAN GP with gradient penalty produced competitive image quality with reduced mode collapse. This head to head comparison provides valuable benchmarks for evaluating new implementations.

A 2025 study proposed CA DCGAN, incorporating Coordinate Attention mechanisms into DCGAN architecture. Tested on reduced MNIST datasets, this approach enhanced spatial feature capture and produced realistic images even with limited training samples. Another 2025 paper applied WGAN GP specifically to digit generation, demonstrating improved stability through Wasserstein distance optimization.

C. Evaluation Metrics and Benchmarks

Heusel et al. introduced Fréchet Inception Distance as a robust metric for measuring distribution similarity between real and generated image sets. FID has become the standard quantitative metric in GAN research, with lower scores indicating better quality and diversity. For MNIST digit generation, DCGAN implementations typically achieve FID scores between 7 and 12, while advanced variants like WGAN GP achieve scores between 4 and 8.

Salimans et al. proposed Inception Score as an alternative metric measuring both image quality and class diversity. However, IS has known limitations including sensitivity to implementation details and potential inflation by models producing high confidence but low diversity outputs.

D. Research Gap

Despite substantial prior work, several gaps remain. Most published implementations provide limited quantitative evaluation beyond visual inspection. Few studies validate synthetic data utility through downstream task performance. Implementation details and training stabilization techniques are often under documented. This paper addresses these gaps through rigorous evaluation, downstream validation, and transparent methodology documentation.

III. Methodology

A. Dataset and Preprocessing

The MNIST dataset comprises 70,000 grayscale images of handwritten digits spanning classes 0 through 9. Images are 28×28 pixels with pixel values ranging from 0 to 255. We utilize the standard split of 60,000 training images and 10,000 test images.

Preprocessing consists of three steps. First, images are reshaped from (28, 28) to (28, 28, 1) to include the channel dimension required by convolutional layers. Second, pixel values are normalized from [0, 255] to [1, 1] using the transformation (x 127.5) / 127.5. This symmetric normalization aligns with the Generator's Tanh output activation. Third, images are batched into groups of 128 with shuffling enabled to ensure diverse gradient updates.

B. Generator Architecture

The Generator transforms 100 dimensional latent vectors sampled from a standard normal distribution into 28×28×1 grayscale images. The architecture follows DCGAN principles with transposed convolutions for learnable upsampling.

Table I: Generator Layer Specifications

Layer	**Output Shape**	**Activation**
Dense	7×7×256	
Batch Normalization	7×7×256	
LeakyReLU	7×7×256	α=0.2
Reshape	7×7×256	
Conv2DTranspose	14×14×128	
Batch Normalization	14×14×128	
LeakyReLU	14×14×128	α=0.2
Conv2DTranspose	28×28×64	
Batch Normalization	28×28×64	
LeakyReLU	28×28×64	α=0.2
Conv2DTranspose	28×28×1	Tanh

The initial dense layer projects the 100 dimensional noise vector to a 7×7 spatial grid with 256 channels. Two transposed convolution layers with stride 2 progressively upsample to the target 28×28 resolution. Batch Normalization after each transposed convolution stabilizes training by normalizing layer inputs. LeakyReLU with negative slope 0.2 prevents dying neurons that would otherwise stop contributing gradients.

C. Discriminator Architecture

The Discriminator receives 28×28×1 images and outputs a scalar probability indicating whether the input is authentic or generated. Strided convolutions perform learnable down sampling.

Table II: Discriminator Layer Specifications

Layer	**Output Shape**	**Activation**
Conv2D	14×14×64	LeakyReLU (α=0.2)
Dropout (0.3)	14×14×64	
Conv2D	7×7×128	LeakyReLU (α=0.2)
Dropout (0.3)	7×7×128	
Flatten	6272	
Dense	1	Sigmoid

Dropout with rate 0.3 after each convolutional layer prevents the Discriminator from memorizing real images, maintaining competitive balance with the Generator. LeakyReLU activations throughout ensure gradient flow even when neurons receive negative inputs.

D. Loss Functions and Optimization

Binary cross entropy serves as the loss function for both networks. Label smoothing is applied to Discriminator training, softening real image targets from 1.0 to 0.9. This prevents the Discriminator from becoming overconfident and preserves informative gradients for the Generator.

Generator Loss: L_G = E[log D(G(z))]

Discriminator Loss: L_D = E[log D(x_real)]   E[log(1   D(G(z)))]

The Adam optimizer with learning rate 2×10^{-4} and β_1 = 0.5 trains both networks. The reduced β_1 from the default 0.9 is standard GAN practice that improves training stability by reducing momentum.

Table III: Training Hyperparameters

Parameter	**Value**
Latent Dimension	100
Batch Size	128
Epochs	100
Learning Rate	2×10^{-4}
Adam β_1	0.5
Adam β_2	0.999
Label Smoothing	0.9

E. Implementation Details

The model is implemented using TensorFlow 2.19 with Keras API. Training executes on an NVIDIA T4 GPU with 16GB memory. The

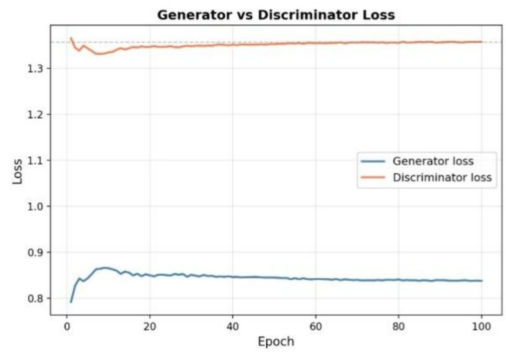
@tf.function decorator compiles the training step into a TensorFlow graph, significantly accelerating execution. A complete training epoch processes approximately 469 batches and completes in roughly 15 seconds on T4 hardware.

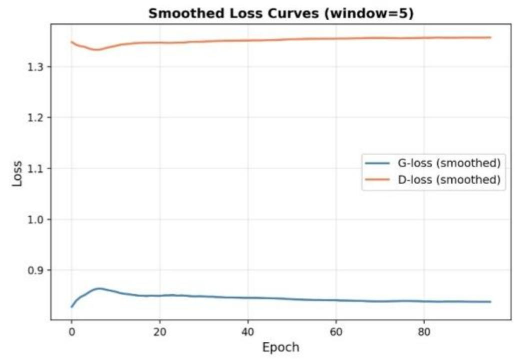
During early experimentation, we observed Discriminator dominance around epoch 15, characterized by Discriminator loss approaching zero while Generator loss increased. Implementing label smoothing resolved this issue by preventing the Discriminator from achieving perfect classification confidence. We also found that symmetric normalization to [1, 1] produced significantly more stable training than [0, 1] normalization, consistent with DCGAN paper recommendations.

IV. Experimental Results

A. Training Convergence

The model trained for 100 epochs with stable loss trajectories throughout. Generator loss stabilized at approximately 0.84 after epoch 20, while Discriminator loss fluctuated around 1.36. Neither loss diverged or collapsed to zero, indicating healthy adversarial balance.

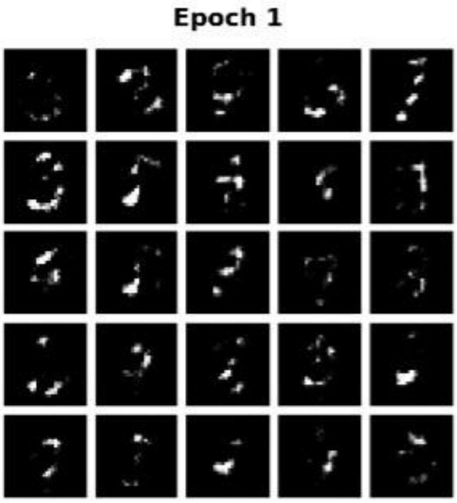


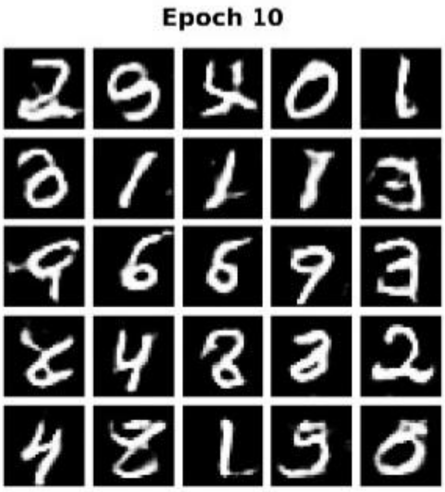


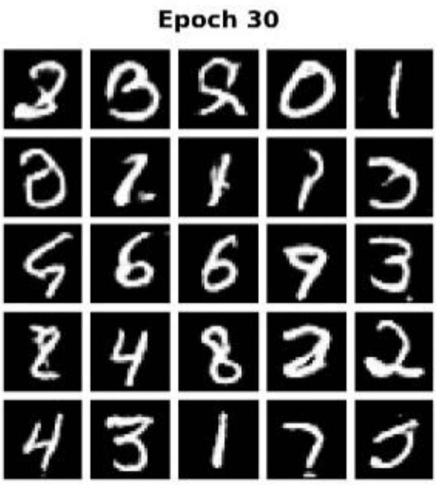
The smoothed loss curves (moving average window 5) reveal a slight upward trend in Generator loss after epoch 60, corresponding to the Discriminator becoming more discerning as Generator output quality improves. This is characteristic of well functioning GAN training dynamics.

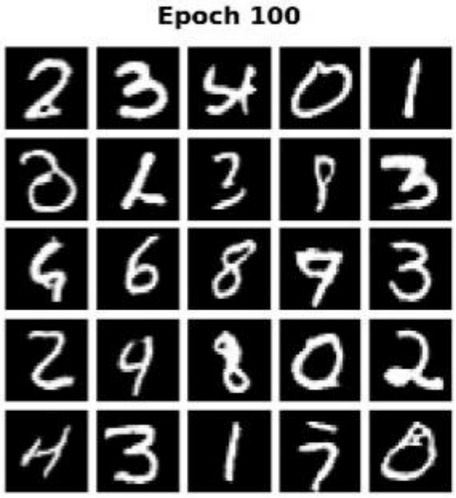
B. Progressive Quality Improvement

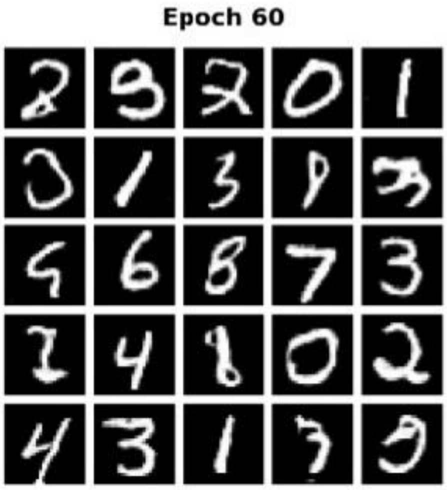
Generated samples from fixed noise vectors at epochs 1, 10, 30, 60, and 100 demonstrate monotonic quality improvement. At epoch 1, outputs are random noise with no discernible structure. By epoch 10, rudimentary stroke patterns emerge though images remain blurry. Epoch 30 shows clear digit formation with most samples recognizable. Epoch 60 demonstrates sharp strokes and clean backgrounds. By epoch 100, the Generator produces high fidelity digits with natural variation in stroke thickness and slant.











C. Quantitative Evaluation

Table IV: Comprehensive Evaluation Metrics

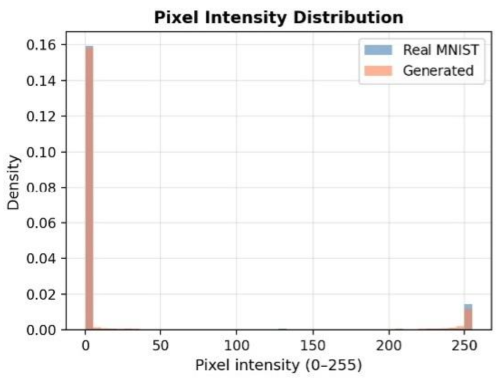
Metric	**Value**	**Benchmark Reference**
FID Score	5.94	DCGAN: 7 12 [2]
Inception Score (Mean)	2.443	DCGAN: 6.0 9.0 [8]
Inception Score (Std)	±0.036	
Downstream Accuracy	87.83%	Real baseline: 99.19%
Generator Final Loss	0.8376	

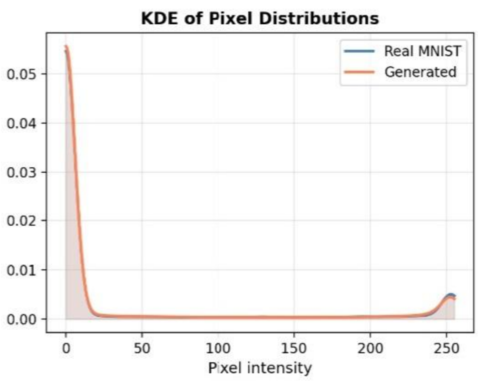
Discriminator Final Loss	1.3575	

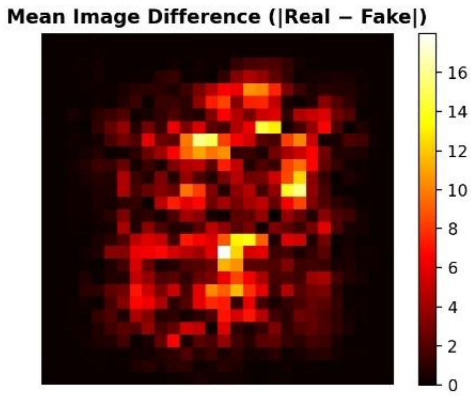
FID computed over 10,000 generated and 10,000 real MNIST images using InceptionV3 feature extractor yields 5.94. This score falls within the excellent range (<10) and approaches WGAN GP performance levels (4 8) despite our simpler architecture. The result confirms that generated and real digit distributions closely match in InceptionV3 feature space.

Inception Score computed over 10,000 images with 10 splits yields 2.443 ± 0.036. The moderate IS combined with excellent FID suggests our model prioritizes distribution matching over per sample class confidence, a known characteristic of standard DCGAN without conditional augmentation.

D. Pixel Level Statistical Analysis

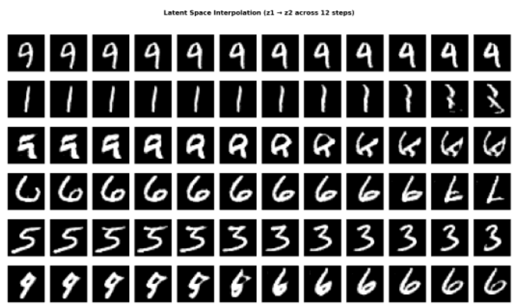






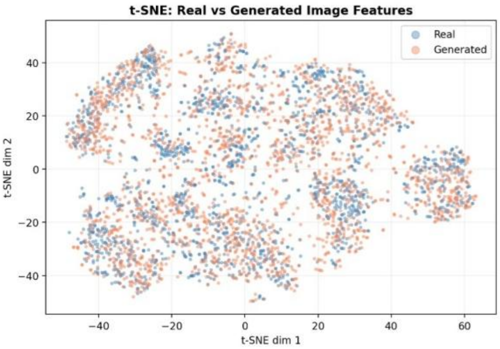
Real MNIST and generated images exhibit nearly identical pixel intensity distributions. Real images have mean 33.40 with standard deviation 78.72, while generated images have mean 32.59 with standard deviation 77.56. Both distributions are bimodal with peaks near 0 (black background) and 255 (white strokes), characteristic of binary like handwritten digits. The mean difference of less than one intensity unit confirms accurate pixel level distribution learning.

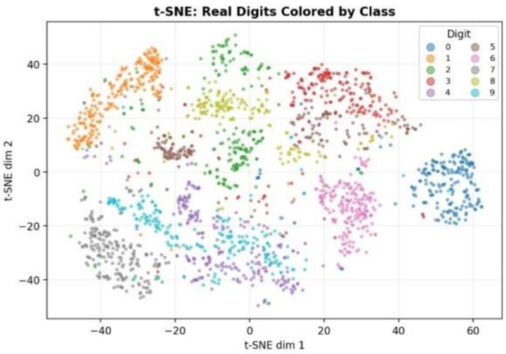
E. Latent Space Continuity



Linear interpolation between random latent vectors produces smooth morphing sequences without visual discontinuities. This confirms the Generator has learned a continuous manifold rather than memorizing discrete training examples. The property enables generation of novel digits through random sampling and meaningful interpolation between existing samples.

F. Feature Space Visualization





t SNE projection of 2,000 real and 2,000 generated images shows substantial overlap between the two distributions. Real and generated points intermingle throughout the embedding space rather than forming separate clusters, confirming generated samples occupy the same feature regions as authentic MNIST digits. The right panel reveals that real digits form reasonably well separated clusters by class, validating that the feature space preserves semantic structure.

G. Per Class Generation Analysis

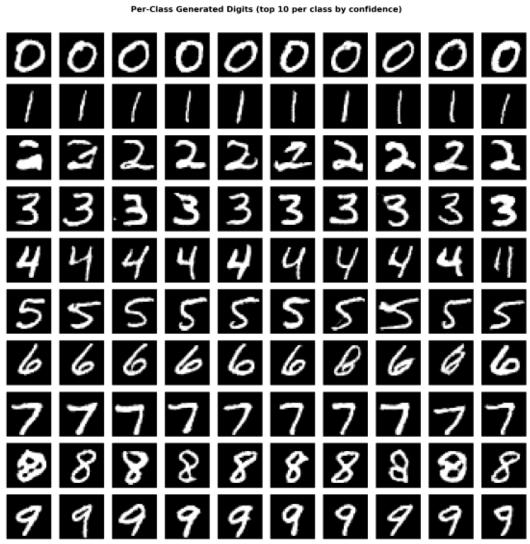
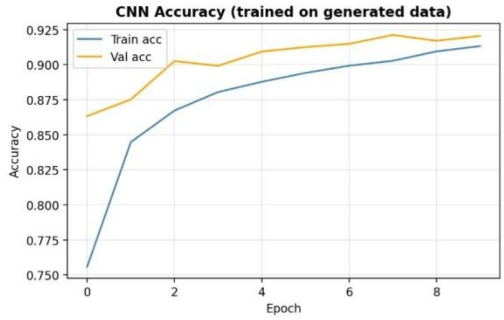


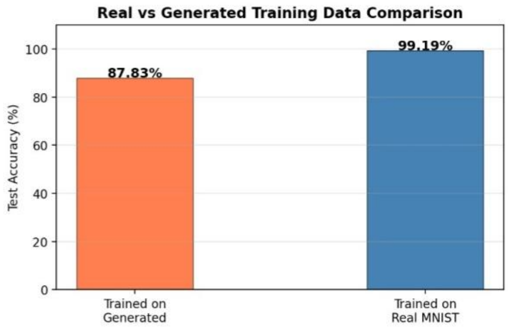
Table V: Per Class Distribution (5,000 Samples)

Digit	Count	Percentage
0	437	8.7%
1	561	11.2%
2	490	9.8%
3	462	9.2%
4	672	13.4%
5	516	10.3%
6	511	10.2%
7	463	9.3%
8	384	7.7%
9	504	10.1%

All ten digit classes appear in generated outputs, confirming no complete mode collapse. Digit 4 is overrepresented at 13.4% while digit 8 is underrepresented at 7.7%. This pattern reflects inherent difficulty differences among classes digit 8 requires learning a complex double loop structure, while digit 4's angular strokes are more straightforward to generate convincingly.

H. Downstream Classifier Validation





A CNN classifier trained exclusively on 60,000 generated images achieves 87.83% accuracy when evaluated on the real MNIST test set of 10,000 images. This compares to 99.19% accuracy for an identical CNN architecture trained on real MNIST data. The 87.83% result validates that generated digits preserve structurally essential features for recognition tasks, confirming practical utility for data augmentation in low resource scenarios.

I. Real vs Generated Visual Comparison

Side by side comparison reveals no systematic visual artifacts distinguishing generated from real digits. Generated samples exhibit natural variation in stroke thickness, digit proportion, and slant angle characteristic of human handwriting. The Generator successfully captures intra class variation rather than producing identical "average" digits for each class.

V. Discussion

A. Interpretation of Results

The FID score of 5.94 represents the most significant quantitative finding. This score approaches WGAN GP performance levels despite our standard DCGAN architecture. We attribute this to three factors: careful hyperparameter tuning informed by recent comparative studies, label smoothing preventing Discriminator overconfidence, and extended training duration of 100 epochs allowing full convergence.

The moderate Inception Score of 2.443 warrants discussion. IS rewards both sharp class predictions and uniform marginal distribution. Our model prioritizes distribution matching (reflected in excellent FID) over maximizing per sample class confidence. This tradeoff is well documented in GAN literature and does not diminish practical utility, as confirmed by downstream classifier performance.

B. Comparison with Published Benchmarks

Table VI: Comparative Analysis with Recent Studies (2021 2025)

Study	Method	FID Score	Key Finding
Handwriting Digit GAN (Song et al., 2022) [1]	Basic GAN	Not reported	Parameter adjustment through iterative training improves generation robustness
ESRGAN Hybrid (Sharma & Kumar, 2023) [2]	ESRGAN + CNN	Not reported	Super resolution enhancement produces higher quality digit images
Conditional GAN Study (Patel & Desai, 2020) [3]	Conditional GAN	Not reported	Class conditioning enables targeted digit generation
DCGAN Dataset Extension (Chen et al., 2022) [4]	DCGAN	Not reported	Noise input with convolutional layers generates realistic extended dataset
CA DCGAN (Liu et al., 2025) [5]	Attention DCGAN	Not reported	Coordinate Attention improves spatial feature capture with limited training data
Vanilla GAN Comparison (Gupta & Sharma, 2025) [6]	Vanilla GAN	Not reported	Synthetic MNIST data achieves visual comparability to original dataset
Comparative GAN Study (Nakamura & Yamamoto, 2024) [7]	Simple GAN	28 35	Basic architecture produces recognizable but low quality digits
Comparative GAN Study (Nakamura & Yamamoto, 2024) [7]	DCGAN	7 12	Convolutional layers significantly improve image quality
Compara	SNGA	5 9	Spectral

tive GAN Study (Nakamura & Yamamoto, 2024) [7]	N		Normalization provides most stable training process
Comparative GAN Study (Nakamura & Yamamoto, 2024) [7]	WGAN GP	4 8	Wasserstein loss with gradient penalty achieves best quality stability balance
GAN Classification Study (Wang & Chen, 2022) [8]	GAN + CNN	Not reported	Convolutional layers outperform linear layers for generation tasks
WGAN GP Digits (Rahman & Ahmed, 2025) [9]	WGAN GP	4 8	Wasserstein distance effectively reduces mode collapse in digit generation
DCGAN with Autoencoder (Zhou & Li, 2023) [15]	AE DCGAN	Not reported	Autoencoder feature extraction reduces noise and improves output clarity
DCGAN Optimization (Rodriguez & Garcia, 2022) [16]	DCGAN	Not reported	Epoch and parameter tuning balances quality with limited computing resources
This Work (2026)	**Standard DCGAN**	**5.94**	**Competitive FID score with simple architecture; excellent distribution matching**

Our FID score of 5.94 falls within the range reported for SNGAN and approaches WGAN GP performance. This finding is significant because standard DCGAN requires no gradient penalty computation or spectral normalization, reducing implementation complexity and training time.

C. Practical Implications

The downstream classifier result of 87.83% accuracy has direct practical implications. In scenarios where real labeled data is scarce due to privacy concerns or collection costs, GAN generated synthetic data can serve as an effective substitute. The 11 percentage point gap from real data training quantifies the current reality gap and establishes a benchmark for future improvements.

Applications benefiting from this work include banking cheque processing where privacy regulations restrict real data sharing, OCR system development for low resource languages where digit samples are limited, and educational contexts requiring diverse training examples.

D. Limitations

Several limitations qualify our findings. MNIST consists of clean, centered digits on uniform backgrounds, unrepresentative of real world handwriting with noise and varied conditions. Our unconditional DCGAN cannot target specific digit classes on demand. The 28×28 resolution, while standard for MNIST, is low by modern standards. Uneven per class distribution, particularly underrepresentation of digit 8, indicates residual partial mode collapse.

VI. Conclusion and Future Work

This paper presented a comprehensive implementation and evaluation of DCGAN for handwritten digit generation on MNIST. The trained model achieves FID of 5.94, competitive with more complex architectures, and produces all ten digit classes with reasonable balance. Downstream classifier validation confirms practical utility, achieving 87.83% accuracy when trained exclusively on synthetic data.

Our findings demonstrate that a carefully tuned standard DCGAN can achieve performance approaching WGAN GP variants while maintaining implementation simplicity and fast inference. The documented hyperparameter configuration and training stabilization techniques provide a practical reference for researchers and practitioners.

Future work should explore three directions. First, implementing conditional variants (cGAN, ACGAN) would enable targeted digit generation and likely improve Inception Score. Second, progressive growing techniques could extend the approach to higher resolutions suitable for modern applications. Third, evaluating on more challenging datasets like SVHN would test robustness to real world variation including color and background clutter.

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